ADDITIONAL MIDTERM 02 REVIEW PROBLEMS – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

⚠ Caution

These problems aren't necessarily indicative of the exact type of problems that will/may appear on the actual exam.

QUESTION 1: AI Detection

It's sometimes stated that Generative AI produces text that is overly verbose. But do AI detectors agree? In particular, Jacinta wants to determine whether a particular AI detector is more likely to return a verdict of "written by AI" for longer pieces of writing than shorter pieces of writing. To that end, she collects 200 pieces of writing from her friends (some of whom used AI to generate their pieces, and some of whom did not) and records whether the piece is "short" (500 words or less; a "**success**") or "long" (more than 500 words; a "**failure**"). She feeds each piece into the AI detector and records whether it returns a verdict of "written by AI" (a "**success**") or "not written by AI" (a "**failure**"). Whenever necessary, use the order (Success — Failure).

Among the 110 short pieces, the detector returned a verdict of "written by AI" for 30 of them. Among the 90 long pieces, the detector returned a verdict of "written by AI" for 75 of them.

(a) Fill in the following two-way table.

	Short	Long	Total
Verdict of AI	**30**	**75**	**105**
Verdict of No AI	**80**	**15**	**95**
Total	**110**	**90**	**200**

(b) Which of the following options give the correct definitions for the explanatory and response variables, and their classification as either categorical or quantitative?

√ **Explanatory: "length"; categorical (with levels "short" and "long")**
Response: "verdict"; categorical (with levels "written by AI" and "not written by AI")

◯ **Explanatory:** "length"; quantitative (with units "number of words")
Response: "verdict"; categorical (with levels "written by AI" and "not written by AI")

(c) Write down the null and alternative hypotheses in symbols and math only, but make sure to define any symbols you use.

Solution: Define:

- π_{Short} : the population proportion of all short works for which the detector would return a verdict of "AI"

- π_{Long} : the population proportion of all long works for which the detector would return a verdict of "AI"

Then our null and alternative hypotheses can be written as

$$\begin{bmatrix} H_0: & \pi_{\text{Short}} = \pi_{\text{Long}} \\ H_0: & \pi_{\text{Short}} < \pi_{\text{Long}} \end{bmatrix}$$

or, equivalently,

$$\begin{bmatrix} H_0: & \pi_{\text{Short}} - \pi_{\text{Long}} = 0 \\ H_0: & \pi_{\text{Short}} - \pi_{\text{Long}} < 0 \end{bmatrix}$$

Notice the sign in the alternative - this comes from the way the research question is phrased. It says "Jacinta wants to determine whether a particular AI detector is more likely to return a verdict of 'written by AI' for longer pieces of writing than shorter pieces of writing."

(d) Are the validity to conduct a theory-based approach to inference met? Explain.

Solution: **Yes**, since all four interior cells of the two-way table have values that are at least 10. In particular:

- The number of short pieces with a verdict of "AI" is 30 (which is at least 10)
- The number of long pieces with a verdict of "AI" is 75 (which is at least 10)
- The number of short pieces with a verdict of "no AI" is 80 (which is at least 10)
- The number of long pieces with a verdict of "no AI" is 15 (which is at least 10)

Remember: part of showing your work is being specific with your counts!

(e) Construct a 95% confidence interval for the parameter of interest. Show your work!

Solution: There's a couple of things to note here. First: the problem statement tells us to use the order of (Success − Failure) whenever necessary; this means our parameter of interest is going to be $\pi_{\text{Short}} - \pi_{\text{Long}}$. This gives us:

- $\hat{p}_1 = \hat{p}_{\text{Short}} = \frac{30}{110} \approx 0.2727$ and $n_1 = 110$
- $\hat{p}_2 = \hat{p}_{\text{Long}} = \frac{75}{90} \approx 0.8333$ and $n_2 = 90$

This allows us to calculate the standard error:

$$\begin{aligned} \text{SE}(\hat{p}_1 - \hat{p}_2) &= \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(2-\hat{p}_2)}{n_2}} \\ &= \sqrt{\frac{0.2727(1-0.2727)}{110} + \frac{0.8333(1-0.8333)}{90}} \\ &= \sqrt{0.001803 + 0.001544} = \sqrt{0.003347} \approx 0.05785 \end{aligned}$$

So, our confidence interval then takes the form

$$(\hat{p}_1 - \hat{p}_2) \pm 2 \cdot \text{SE}(\hat{p}_1 - \hat{p}_2) = (0.2727 - 0.8333) \pm 2 \cdot 0.05785$$
$$= -0.5606 \pm 0.1157 = \mathbf{(-0.6763\,,\,-0.4449)}$$

(f) A 90% confidence interval for the parameter of interest is (–0.656 , –0.465). Interpret this interval.

Solution: We are 90% confident that the population proportion of all SHORT works for which the detector returns a verdict of "AI" is between 46.5 and 65.6 percentage points LOWER than the population proportion of all LONG works for which the detector returns a verdict of "AI."

Remember: when interpreting an interval involving TWO parameters (or, more specifically, the DIFFERENCE in two parameters) it's important to include a statement of which (if either) is LARGER/HIGHER or SMALLER/LOWER.

QUESTION 2: Life Expectancies (Again!)

Do countries in Asia have a significantly different life expectancy than countries in Europe? To answer this question, we'll explore a subset of data from the *World Bank*[1], collected in 2024. In particular, we'll explore a dataset containing information from 70 randomly-selected countries from across these two continents.

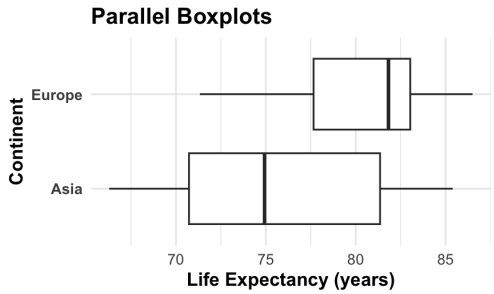


Figure 1: Parallel Boxplots of the Data.

Asia	Europe
$n_{\text{Asia}} = 32$	$n_{\text{Europe}} = 38$
$\bar{x}_{\text{Asia}} = 75.882$ yrs.	$\bar{x}_{\text{Europe}} = 79.767$ yrs.
$s_{\text{Asia}} = 6.047$ yrs.	$s_{\text{Europe}} = 3.413$ yrs

Table 1: Summary Statistics

Figure 1 displays parallel boxplots of the data, and Table 1 displays some summary statistics of the data. Additionally, whenever necessary, use the order (Asia — Europe).

(a) What is one observational unit in this dataset?

Solution: One observational unit is one country

(b) What is the sample median of life expectancies among the Asian countries included in the dataset?

Solution: 75 years, as indicated by the vertical bar in the middle of the "Asia" boxplot

(c) What is the sample IQR of life expectancies among the European countries included in the dataset?

Solution: 83 – 77.5 $\approx$ **5.5 years**, which corresponds to the width of the "Europe" boxplot.

(d) Are the validity to conduct a theory-based approach to inference met? Explain.

[1] https://data.worldbank.org/

Solution: We have at least 20 observations in each group (32 in Asia, 38 in Europe), and the sample distributions (provided in the form of parallel boxplots) don't appear too skewed. So, **yes**; the validity conditions are met.

Note: for this particular dataset, I would have also accepted a "no" answer because the "Europe" boxplot could be interpreted as exhibiting a potential left-skew (since the median is leaning a little closer to the third quartile than the first quartile). However, again, I'd mainly be looking for your *explanations* here (on the quiz, I bypassed this subjectivity by making the question multiple choice). As a quick reminder: when identifying skew on a boxplot, look at the comparative lengths of the whiskers as well as the position of the median.

(e) Calculate $\text{SE}(\bar{x}_1 - \bar{x}_2)$.

Solution:

$$\text{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{6.047^2}{32} + \frac{3.413^2}{38}} = \sqrt{1.44924} \approx \mathbf{1.2038}$$

(f) Calculate the value of the standardized statistic.

Solution:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{75.882 - 79.767}{1.44924} = \frac{-3.885}{1.44924} \approx \mathbf{-2.681}$$

(g) Liam, a (fictional) STAT 1110 student, has used the provided data and obtained a theory-based p-value of around 0.006. As a part of a Lecture Activity, Liam was asked to **interpret** (not "evaluate") his p-value . To that end, he has provided the following **interpretation**:

> The p-value of 0.006 provides very strong evidence against the null, indicating a statistically significant difference in the average life expectancy of countries in Asia as compared to countries in Europe.

Has Liam correctly interpreted this p-value? Explain.

Solution: Though Liam's statement is correct and consistent, it is not an interpretation of the p-value; it is an **evaluation**. Remember: interpreting a p-value means telling the reader what that value actually represents - i.e. restating the definition of the p-value, but in the context of the problem.

Here's a little more detail on how I'd phrase an interpretation of this p-value. First, remember the working definition I've been using:

> The p-value is the probability of observing something as or more extreme as what we observed under the null.

We now need to make this more specific, and tailor-made to the context of the problem.

- "observing something–" what statistic? (Answer: a difference in sample mean life expectancies, using the order Asia — Europe.)

- "what we observed–" what value? (Answer: our observed difference in sample mean life expectancies; -3.885 years.)
- "as or more extreme–" in which direction? (Answer: either direction, since we're using a two-sided alternative.)
- "under the null–" what does that mean? (Answer: assuming there is no difference in the population average life expectancies of Asian and European countries.)

So, putting everything together:

> 0.006 is the probability of observing a difference in sample mean life expectancies (using the order Asia — Europe) of –3.885 years or more extreme in either direction, assuming the population life expectancies of all Asian countries is the same as the population life expectancies of all European countries.

I encourage you to take a look at these slides as well.